# Software Test Plan (STP) — Online Examination Portal

**Course:** UE24CS341A — Software Engineering, PES University, Dept. of CSE
**Project:** Online Examination System with Multiple-Choice Questions and Evaluation
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**Version:** 5.2 · **Date:** 2026-11-12
**Status:** Executed against build 2d74453 on 3 Nov 2026: 118 of 119 cases Pass, ST_013 failing on Defect 1 (FR-028). Defect 1 fixed and ST_013 re-run on 11 Nov (Deviations D-15): **119 of 119 cases Pass**

## Revision history

| Version | Date | Author | Change summary |
|---|---|---|---|
| M1 | 2026-09-06 | Kshitij | 118 test cases across 13 use cases (68 UT / 37 IT / 13 ST), written to the course test case template |
| 3.0 | 2026-10-15 | Kshitij | Rebuilt to the Part-2 deliverables |
| 3.1 | 2026-11-01 | Kshitij | IT_038 added for NFR-012: 119 cases |
| 4.0 | 2026-11-03 | Kshitij | Actual Result and Test Result recorded for all 119 cases; Deliverable 3 case corrections |
| 4.1 | 2026-11-03 | Kshitij | Arranged to the course's test plan outline; §5.1 Security Validation added; ST_013 Fail on Defect 1 |
| 4.2 | 2026-11-11 | Kshitij | ST_013 Pass after the Defect 1 fix |
| 5.0 | 2026-11-11 | Kshitij | Arranged to the course STP template (§1–§15): test items, features (not) to be tested, entry criteria, schedule, risks, assumptions, metrics and approvals added; test cases moved to Appendix A unchanged |
| 5.1 | 2026-11-12 | Kshitij | 200-user load test run for NFR-001 and NFR-003 and moved from §4 to §3; load-test environment (§6.2) and risk (§10) added (Deviations D-18) |
| 5.2 | 2026-11-12 | Kshitij | CR-02: UT_033, UT_034, UT_035, IT_016, IT_017, ST_005, ST_006 and ST_007 rewritten so Faculty sends the paper and the Administrator schedules it; all re-run and passing; four CR-02 tests added to the suite (Deviations D-20) |

## 1. Introduction

### 1.1 Purpose

This document defines the test plan for the Online Examination Portal: objectives, scope, strategy, resources, schedule and responsibilities. It also records the result of executing every test case.

**Objectives.**

- Every functional requirement is verified by at least one passing test case (NFR-007).

- Every test case that can be automated is, and runs on every push and pull request in CI.
- No release with a failing test: a red CI run blocks the merge to `main`.
- Security requirements are verified both by automated tests and by manual checks against the running system (§5.1).

### 1.2 Scope

Testing covers all 30 functional requirements across the 13 use cases, and the 11 non-functional requirements that can be checked on one machine (§3). Coverage is **8–10 test cases per use case**, **119 cases** in all, at Unit, Integration and System level.

The features not tested, and why, are in §4.

### 1.3 References

- Software Requirements Specification v4.0 — `docs/SRS.pdf` (requirement IDs, acceptance criteria, RTM).
- Software Architecture and Design Specification v3.0 — `docs/SAD.pdf`.
- Use Case Diagram and Use Case Flow — `docs/Use_Case_Diagram.pdf`, `docs/Use_Case_Flow.pdf`.
- System Validation Report, Unit Test Report, Code and Branch Coverage Reports — `docs/`.
- Deviations register — `docs/Deviations.pdf`.
- The course test case template, `Test Plan Document.docx`, which sets the columns of Appendix A.

### 1.4 Definitions

| Term | Meaning |
|---|---|
| UT / IT / ST | Unit, integration and system test case (§5) |
| FR / NFR / UC | Requirement and use case IDs of the SRS |
| SO | Security objective (SRS §5.1.1) |
| RTM | Requirements Traceability Matrix (SRS §8) |
| CI | Continuous integration — GitHub Actions running the suite on every push and pull request |
| S1 … S4 | Defect severity (§5.4) |
| Fixture | A named, seeded test account or record (§6.3) |
| Frozen build | Commit `2d74453`, the build every case was executed against |

## 2. Test Items

The build under test is commit `2d74453` of `main` (end of Sprint 2), plus the Defect 1 fix of 11 Nov for the ST_013 re-run. Each Django app is a test item:

- `identity` — registration, sign-in, roles, lockout, enrollment, eligibility, DPDP consent and erasure
- `questionbank` — questions, options, tags, versioning

- `examsetup` — examination details and policy, question selection, publish, schedule
- `attempts` — dashboard, start, timer, paper, palette, autosave, resume, submit and auto-submit
- `evaluation` — scoring, percentage, pass/fail
- `results` — withholding, publish and unpublish, scorecard
- `integrity` — focus-loss events, integrity and audit views
- `repository` — models and the append-only audit trail
- `portal` — settings, role middleware, error pages, traceability matrix, page templates

## 3. Features to be Tested

Features mapped to SRS requirement IDs:

| Feature | Requirements | Use cases |
|---|---|---|
| Registration, sign-in and role-based access | FR-001, FR-002, FR-003 | UC-01, UC-02 |
| Courses and enrollment | FR-004, FR-005, FR-006 | UC-03, UC-04 |
| Question bank | FR-007, FR-008, FR-009 | UC-05 |
| Examination authoring and publication | FR-010, FR-011, FR-012 | UC-06 |
| Scheduling and the Student dashboard | FR-013, FR-014 | UC-07 |
| Starting an attempt, timer and palette | FR-015 … FR-018 | UC-08 |
| Autosave, resume and auto-submission | FR-019, FR-020, FR-021 | UC-09, UC-10 |
| Automatic evaluation | FR-022, FR-023, FR-024 | UC-11 |
| Results and scorecard | FR-025, FR-026, FR-027 | UC-12 |
| Integrity events, answer-key protection and audit trail | FR-028, FR-029, FR-030 | UC-13 |
| Response time and capacity: 200 attempts, 50,000 questions | NFR-001, NFR-003 | — |
| Evaluation time ≤ 5 s | NFR-002 | — |
| Recovery after restart | NFR-004 | — |
| Browsers and 360 px width | NFR-005 | — |
| Every FR has a test case | NFR-007 | — |
| DPDP consent and erasure | NFR-008 | — |
| WCAG 2.1 AA | NFR-009 | — |
| Configurable exam policy | NFR-010 | — |
| Password hashing, lockout, CSRF/ORM/error pages, secrets | NFR-011 … NFR-014 | — |

## 4. Features Not to be Tested

- **NFR-006** — a process requirement, checked on the GitHub history rather than by a test.
- **Django, the browsers and SQLite themselves** — assumed tested by their maintainers.

- **Nginx and HTTPS configuration** — no deployment server exists for the course. The load test runs Gunicorn on a workstation instead (§6.2).
- **Everything the SRS puts out of scope** — proctoring, descriptive answers, payments, mobile apps.

## 5. Test Approach / Strategy

Tests are written with the feature, not after it. Whoever builds a requirement writes its tests in the same pull request.

Each automated test is named after its case ID (`test_UT_001_…`), so a failing test points straight at the case and the requirement behind it. `portal/test_traceability.py` fails the build if a case in this plan has no test.

**Levels.** Case IDs follow the template's guidance (*"follow some convention like `UT_01`"*):

| Prefix | Test level | What it exercises | How it runs |
|---|---|---|---|
| UT_nnn | **Unit test** | One function, validator or rule in isolation — field validation, a scoring calculation, a state transition | Django test runner |
| IT_nnn | **Integration test** | Two or more components together — view to service to database, page to API | Django test client |
| ST_nnn | **System test** | An end-to-end flow through the running system as a real actor would drive it | Django test client in CI; re-run by hand in real browsers for the System Validation Report |

Numbering is a single running sequence per prefix, so an ID is never reused if a case is later inserted or retired. Acceptance is the system level re-run in real browsers, recorded in the System Validation Report.

**Types.**

- **Functional** — every case in Appendix A.
- **Regression** — the whole suite on every push and pull request in CI.
- **Performance** — evaluation time (NFR-002, IT_030); the 200-user Locust load test (NFR-001, NFR-003).
- **Recovery** — server restart during an attempt (NFR-004).
- **Security** — §5.1.
- **Usability and accessibility** — axe audit and a keyboard-only attempt (NFR-009).
- **Portability** — Chrome, Edge and Firefox, and 360 px width (NFR-005).

### 5.1 Security Validation

How each security requirement is validated, by whom and when. Automated checks run on every push in CI; the manual checks are run against the live system for the System Validation Report.

**Security objectives** (from `docs/SRS.pdf` §5.1.1):

- **SO-1 Confidentiality of the examination** — no Student sees the answer key, another Student's attempt, or a result before it is published.
- **SO-2 Integrity of attempts and scores** — time limits, scores and the audit trail cannot be altered from the browser.
- **SO-3 Protection of accounts and secrets** — credentials and keys stay safe even if the database or the repository is exposed.

| Requirement | Objective | What must be confirmed | Validation method | Test cases / evidence | Owner | Timing | Tools |
|---|---|---|---|---|---|---|---|
| FR-003 Role-based access | SO-1 | Every protected URL refuses other roles (403) and anonymous users (302 / 401) | Automated integration tests; manual URL tampering | UT_011, IT_004, IT_005, IT_006, ST_002 | Kshitij | Sprint 1 CI; Deliverable 3 validation | Django test client, Playwright |
| FR-006, FR-025 Own data only | SO-1 | Another course's exam and another Student's result return 403 | Automated tests; manual URL tampering | IT_009–IT_011, IT_032, IT_033, ST_004, ST_012 | Komal | Sprint 1–2 CI; validation | Django test client, Playwright |
| FR-029 No answer key in payloads | SO-1 | No page or JSON reply of an attempt carries `is_correct` or any key marker | Automated payload test; manual inspection of every payload the browser received | IT_035, ST_013 | Kartik | Sprint 1 CI; validation | Django test client, Playwright response capture |
| FR-017, FR-020 Server-side timing | SO-2 | A changed client clock changes nothing; expiry submits at start + duration exactly | Automated tests; manual 3-minute expiry run | UT_042, IT_027, ST_010 | Kartik, Kshitij | Sprint 1 CI; validation | Django test client, Playwright |
| FR-030 Append-only audit | SO-2 | Every change writes one entry; no screen or endpoint edits or deletes one | Automated tests; manual reconciliation of the audit log after one exam cycle | UT_068, IT_036, IT_037, ST_013 | Komal | Sprint 2 CI; validation | Django test client, Playwright |
| NFR-011 Password hashing | SO-3 | Stored passwords are `pbkdf2_sha256$…` only, never plain text | Automated assertion; manual read of the password column | `identity/test_auth.py`; validation check NFR-011 | Kartik | Sprint 1 CI; validation | Django test runner, ORM query |
| NFR-012 Account lockout | SO-3 | The 6th sign-in after 5 failures is refused even with the right password; it succeeds after 15 minutes | Automated test | IT_038 (`identity/test_lockout.py`) | Kshitij (D-09) | Sprint 2 CI | Django test client |
| NFR-013 CSRF, ORM-only, generic errors | SO-2, SO-3 | POST without a token → 403; no raw SQL; a forced error shows no stack trace with DEBUG off | Automated test; static code search; manual forced 404 | `portal/test_security.py`; code search for `.raw(`, `cursor(`, `.extra(`; validation check NFR-013 | Komal | Sprint 1 CI; validation | Django test client, `git grep`, Playwright |
| NFR-014 Secrets from the environment | SO-3 | No secret in the repository or its history; the app refuses to start without `DJANGO_SECRET_KEY` | Secret scan of the full git history; manual start without the variable | validation check NFR-014 | Komal | Sprint 1; validation | `git log -p` scan, `manage.py check` |

## 5.2 Entry criteria

Execution of the plan starts when:

- the build is frozen (commit 2d74453) so the cases cannot drift while they run;
- CI is green on that commit;
- the fixtures of §6.3 can be seeded into a fresh database;
- every case in Appendix A has an automated test, or is marked manual (IT_022).

### 5.3 Exit criteria

Testing is complete when:

1. every one of the 119 cases has been executed and has a result recorded in this plan;
2. every automated test passes on the frozen build;
3. no S1 or S2 defect is open;
4. statement coverage is at least 80% and branch coverage at least 70% (targets set in 4.1; measured 88.8% and 72.1% on build 2d74453);
5. the System Validation Report records the manual checks of §5.1.

At v5.0 all five criteria are met and no defect is open: Defect 1 (S3) was fixed on 11 Nov and ST_013 passes. NFR-001 and NFR-003 pass the 200-user load test on 12 Nov (System Validation Report §4.1).

### 5.4 Bug triage

| Severity | Meaning | Example | Rule |
|---|---|---|---|
| **S1 Critical** | Wrong score, lost answers, a security control bypassed | An answer key reaching the browser | Blocks the release; fixed before anything else |
| **S2 Major** | A requirement not met, no workaround | Results visible before publication | Fixed in the current sprint |
| **S3 Moderate** | A requirement met only partly, or wrong evidence produced | Extra focus-loss events on question navigation (FR-028, Defect 1 of the System Validation Report) | Fixed in the next sprint; recorded in Deviations |
| **S4 Minor** | Cosmetic, wording, layout | A label without spacing | Fixed when convenient |

A defect is raised as a GitHub issue with its severity and the failing case ID.

## 6. Test Environment

### 6.1 Testing tools

| Tool | Used for |
|---|---|
| Django test runner (`manage.py test`, Python `unittest`) | All 118 automated cases |
| coverage.py 7.16 (`coverage run --branch`) | Code and Branch Coverage reports |
| GitHub Actions (Ubuntu, Python 3.12) | Runs the suite on every push and pull request |
| Playwright 1.64 (Python) with Chrome, Edge and Firefox | IT_022 timing and the System Validation walkthrough |

| Tool | Used for |
|---|---|
| axe-core 4.10 | WCAG 2.1 AA audit of the attempt flow (NFR-009) |
| Locust 2.46 | Load script for NFR-001 / NFR-003 (`loadtest/locustfile.py`) |
| Gunicorn 27 | The application server for the load test, as in SAD deployment |

## 6.2 Hardware, software and data

| | |
|---|---|
| Automated runs | A fresh SQLite test database created by the Django test runner for each run, on Windows 11 (Python 3.10) and in CI (Ubuntu, Python 3.12) |
| System validation | The application under `manage.py runserver` with DEBUG off, on a throwaway SQLite file seeded with the fixtures below (OEP_DB_PATH), driven in real Chrome, Edge and Firefox |
| Load test | Gunicorn, 25 workers, in WSL 2 Ubuntu 22.04 on a 12-thread Windows 11 laptop; Python 3.10, Django 5.2; a fresh copy of a SQLite file seeded by `manage.py seed_bank` (50,000 questions, 200 enrolled students) for every run; Locust on the same machine |
| Client | Windows 11; viewport 1280×800, and 360×800 for NFR-005 |
| Server clock | The host clock, NTP-synchronised; every timing rule reads the server clock only |
| Data | Seeded fixture set — see §6.3 |

## 6.3 Standing test fixtures

Referenced by name in the *Test data* column so precise values are not repeated in every row.

| Fixture | Value |
|---|---|
| STU_A | `stu.a@pesu.pes.edu` / `Passw0rd!A` / roll `PES1UG24CS001`, role Student, active |
| STU_B | `stu.b@pesu.pes.edu` / `Passw0rd!B` / roll `PES1UG24CS002`, role Student, active |
| FAC_A | `fac.a@pes.edu` / `Fac#2026A`, role Faculty, assigned to CRS_X |
| FAC_B | `fac.b@pes.edu` / `Fac#2026B`, role Faculty, assigned to CRS_Y only |
| ADM_A | `admin@pes.edu` / `Adm#2026A`, role Administrator |
| CRS_X | Course UE24CS351A / "Database Management Systems" / Semester 5, active |
| CRS_Y | Course UE24CS342AA5 / "Advanced Virtual Reality" / Semester 5, active |
| EXM_1 | Examination "DBMS Unit Test 1", CRS_X, 60 min, declared total 50, 25 questions, published |
| Q_SC1 | Single-correct MCQ, 4 options, correct = option B, +2 marks, −0.5 negative |

# 7. Test Schedule

| Milestone | Date |
|---|---|
| Test case design: 118 cases for 13 use cases (M1) | 2026-09-06 |
| Test Plan rebuilt for the Part-2 deliverables | 2026-10-15 |
| Sprint 1 — tests written with each feature, CI on every push | 2026-10-19 to 2026-10-23 |
| Sprint 2 — tests written with each feature, CI on every push | 2026-10-26 to 2026-11-01 |

| Milestone | Date |
|---|---|
| IT_038 added; every case keyed to an automated test or marked manual | 2026-11-01 |
| Build frozen at 2d74453 | 2026-11-01 |
| Execution of all 119 cases; Code and Branch Coverage; Unit Test Report | 2026-11-03 |
| System validation in real browsers (System Validation Report v1.0 and v1.1) | 2026-11-02 to 2026-11-03 |
| Defect 1 fixed; ST_013 re-run | 2026-11-11 |
| 200-user load test (NFR-001, NFR-003) | 2026-10-28 (PR #87) and 2026-11-12 |

## 8. Test Deliverables

| Deliverable | Where |
|---|---|
| Test Plan, with Actual Result and Test Result for all 119 cases (this document) | docs/Test_Plan.pdf |
| Test cases (manual and automated) | Appendix A; automated tests in each app's test_*.py |
| Test data | The fixtures of §6.3 |
| Unit Test Report | docs/Unit_Test_Report.pdf |
| System Validation Report, with screenshots | docs/System_Validation_Report.pdf, docs/validation/ |
| Code Coverage Report | docs/Code_Coverage_Report.pdf |
| Branch Coverage Report | docs/Branch_Coverage_Report.pdf |
| Requirements Traceability Matrix in code | portal/traceability.py, checked by portal/test_traceability.py |
| Test execution logs | GitHub Actions, workflow *Tests* |
| Defect reports | GitHub issues; Defect 1 in the System Validation Report |

## 9. Roles and Responsibilities

| Role | Name | Responsibility |
|---|---|---|
| Backend & Data, repository owner | Kshitij G Shettigar | Tests for own FRs; CI workflow; Code and Branch Coverage reports; System Validation Report (reassigned on 3 Nov, docs/Deviations.pdf D-12); reviews and merges every PR |
| Frontend & UX | Kartik Basavaraj Meti | Tests for own FRs; review of own 43 cases for Deliverable 3; browser checks of NFR-005 and NFR-009 |
| QA & Docs | Komal S Sajja | Tests for own FRs; review of own 36 cases; Unit Test Report |

Ownership of each FR follows the Sprint Plan (docs/Sprint_Plan.pdf). Whoever builds a requirement writes its tests in the same pull request.

## 10. Risks and Mitigation

| Risk | Mitigation |
|---|---|
| The Django test client runs no JavaScript, so a defect in the page scripts passes CI | System cases re-run in real Chrome, Edge and Firefox for the System Validation Report; this is how Defect 1 |

| Risk | Mitigation |
|---|---|
| | was found |
| SQLite has one writer: under 200 users, one of four load runs reached p95 3.6 s with 2 lock timeouts | WAL mode, IMMEDIATE transactions and a 20 s busy timeout (PR #87); every run reported, not just the passing ones (Deviations D-18); re-run the load test before any real deployment |
| A case loses its automated test, or a test is keyed to the wrong case | `portal/test_traceability.py` fails the build |
| The build changes while the cases are being executed | Build frozen at `2d74453` for the execution |
| A browser cannot be made to switch tabs under automation | Tab switches simulated with window blur and focus events, stated in the System Validation Report |
| An assertion that depends on random page content (the CSRF token) fails at random | Such assertions rewritten to check the field, not the whole page (PR #106) |

## 11. Assumptions & Dependencies

- The fixtures of §6.3 can be seeded into a fresh database for every run.
- The server clock is synchronised by NTP; every timing case reads the server clock only.
- Current versions of Chrome, Edge and Firefox are available for the browser runs.
- Two decisions remain outstanding with the course instructor, and each leaves specific expected values provisional:

| Open question | Test cases affected | Effect if resolved differently |
|---|---|---|
| Negative-marking convention — a flat deduction per wrong answer, or a fraction of the question's marks | UT_054, UT_056, IT_031 | The expected values assume a **flat** deduction of 0.5. A fractional convention changes each expected total |
| Whether a Pass at exactly the pass mark is inclusive | UT_057 | The expected outcome assumes **inclusive** — 40.0% against a 40% pass mark is a Pass |

## 12. Suspension & Resumption Criteria

Testing is **suspended** when the build will not start, when `main` fails CI, or when an S1 defect is open, because results after it would not be trusted.

It **resumes** once the fix is merged and the full suite is green again on `main`. The suspended cases are then re-run from the start.

## 13. Test Case Management & Traceability

### 13.1 Traceability to the SRS

Every test case description opens with the requirement ID it verifies, so this document feeds the test-case column of the RTM (SRS §8) without a separate mapping table.

Identifiers are the `FR-nnn` IDs of the SRS, which are also the backlog items on the GitHub Project. The same matrix is held in code (`portal/traceability.py`) and checked against the test suite by `portal/test_traceability.py`, so it cannot silently go stale.

## 13.2 Coverage per use case

| Use case | Requirements verified | Unit | Integration | System | Total |
|---|---|---|---|---|---|
| UC-01 Register a Student account | FR-001 | 6 | 2 | 1 | 9 |
| UC-02 Sign in and role-based access | FR-002, FR-003, NFR-012 | 5 | 5 | 1 | 11 |
| UC-03 Manage courses and assign faculty | FR-004 | 5 | 2 | 1 | 8 |
| UC-04 Enroll students and gate visibility | FR-005, FR-006 | 5 | 3 | 1 | 9 |
| UC-05 Author question bank entries | FR-007, FR-008, FR-009 | 6 | 3 | 1 | 10 |
| UC-06 Assemble and publish an examination | FR-010, FR-011, FR-012 | 6 | 2 | 1 | 9 |
| UC-07 Schedule and surface on the dashboard | FR-013, FR-014 | 5 | 2 | 1 | 8 |
| UC-08 Start an attempt and receive the paper | FR-015, FR-016, FR-017, FR-018 | 6 | 3 | 1 | 10 |
| UC-09 Autosave and resume | FR-019, FR-021 | 4 | 4 | 1 | 9 |
| UC-10 Automatic submission on expiry | FR-020 | 4 | 3 | 1 | 8 |
| UC-11 Evaluate automatically | FR-022, FR-023, FR-024 | 6 | 3 | 1 | 10 |
| UC-12 Publish results and scorecard | FR-025, FR-026, FR-027 | 5 | 3 | 1 | 9 |
| UC-13 Integrity events and audit trail | FR-028, FR-029, FR-030 | 5 | 3 | 1 | 9 |
| **Total** | **30 of 30** | **68** | **38** | **13** | **119** |

Every use case carries at least eight test cases, and every one of the 30 functional requirements of `docs/SRS.pdf` is exercised by at least one test case — satisfying NFR-007.

## 13.3 How the result columns are used

Up to v3.1 these two columns were left blank, because no build existed to run the cases against. From v4.0 they hold the results of one execution of the whole plan against the frozen Deliverable 3 build, commit `2d74453`, on 3 Nov 2026:

- **118 cases are automated.** Each automated test is named after its case ID (`test_UT_001_…`), so the *Actual Result* names the test or tests that ran the case and their outcome. The run was `coverage run manage.py test -v 2`: 227 tests, all passing, none skipped. The same run produced the Code Coverage and Branch Coverage reports.
- **IT_022 is manual**, because it measures timing inside a real browser. It was run in Chromium through Playwright against a local server, and its *Actual Result* records the measured times.

A case is **Pass** only when every test keyed to it passed. Hand walkthroughs of the system test cases belong to the separate System Validation Report.

**v4.1:** the System Validation Report re-ran all 13 system cases in real browsers. ST_013 failed there on Defect 1 (FR-028: moving between questions records extra focus-loss events), which the automated test cannot see because the Django test client runs no JavaScript. ST_013 is therefore recorded as **Fail**, and the plan stood at **118 of 119 Pass**.

**v4.2:** Defect 1 was fixed on 11 Nov, after the freeze (Deviations D-15). ST_013 was re-run in Chrome through the same script, and passes: 2 focus-loss events for 2 tab switches, unchanged after answering three questions and submitting. The Django suite still passes (227 tests). The plan stands at **119 of 119 Pass**.

### Changes to the cases in v4.0

These cases were corrected in the Deliverable 3 test-case review, so that each one describes what the application now does and what its test checks:

| Case | Change | Why |
|---|---|---|
| IT_006 | Protected pages redirect to sign-in; only API endpoints answer HTTP 401 | A browser is sent to sign in rather than shown a bare 401 |
| IT_011 | Now direct access to another course's examination, refused with 403 | IT_010 already covers listing; FR-006 also forbids the direct URL |
| UT_052 | Submitted is marked by the submission timestamp, not a SUBMITTED state | The attempt model has no separate state field |
| UT_060 | Adds the refused publication while the window is open | FR-026 allows publication only after the window closes (PR #95) |
| UT_068 | The audit entry also carries the old and new answer key | FR-030 audits answer-key changes (PR #96) |
| IT_036 | Five classes are User, Course, Question, Exam, Attempt; adds actor, exam-field and publication checks | Matches the audit coverage added in PR #96 |
| ST_006 | Notes that scheduling is covered by ST_007 | Publication now comes before scheduling |

## 14. Test Metrics & Reporting

| Metric | Value (v5.0) |
|---|---|
| Test cases planned | 119 (68 UT / 38 IT / 13 ST) |
| Test cases executed | 119 of 119 (100%) |
| Test cases passed / failed | 119 / 0 (100% pass) — ST_013 failed on 3 Nov and passed on its 11 Nov re-run |
| Automated tests in the suite | 231, all passing, none skipped (227 before CR-02 added four) |
| Requirement coverage | 30 of 30 FRs; the 13 NFRs of §3 all pass |
| Statement / branch coverage | 88.8% / 72.1% (targets 80% / 70%) |
| Defects found / fixed / open | 1 / 1 / 0 — Defect 1, severity S3 (FR-028) |
| Defect aging | Defect 1: found 3 Nov, fixed 11 Nov — 8 days |

**Reports.**

- Per push and pull request: the GitHub Actions run of the whole suite.
- Per execution: the Actual Result and Test Result columns of Appendix A.
- Final: the Unit Test, System Validation, Code Coverage and Branch Coverage reports of §8.

## 15. Approvals

| Role | Name | Signature / Date |
|---|---|---|
| Course Coordinator | | |
| Team — Backend & Data | Kshitij G Shettigar | |
| Team — Frontend & UX | Kartik Basavaraj Meti | |

| Role | Name | Signature / Date |
|---|---|---|
| Team — QA & Docs | Komal S Sajja | |

## Appendix A — Test cases by use case

The 119 test cases, in the columns of the course test case template. Each description opens with the requirement it verifies (§13.1).

### UC-01 — Register a Student account

Verifies FR-001.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_001 | User Registration | FR-001 — Register with all four mandatory fields valid | Registration page reachable; no account exists for the email or roll number | 1: Open /register<br>2: Enter full name, institutional email, roll number, password<br>3: Click Register | Name Arjun Rao, email arjun.rao@pesu.pes.edu, roll PES1UG24CS111, password Str0ng#Pass | Account created with role Student; message "Registration successful" | As expected. Automated: test_UT_001_register_with_all_fields_valid (identity.test_auth): ok | **Pass** |
| UT_002 | User Registration | FR-001 — Reject submission with a mandatory field left empty | Registration page reachable | 1: Open /register<br>2: Leave Roll Number empty<br>3: Complete the other three fields<br>4: Click Register | Name Arjun Rao, email arjun.rao@pesu.pes.edu, roll "" (empty), password Str0ng#Pass | Registration rejected; no account created; inline error "Roll number is required" | As expected. Automated: test_UT_002_roll_number_is_required (identity.test_auth): ok | **Pass** |
| UT_003 | User Registration | FR-001 — Reject a non-institutional email domain | Registration page reachable; accepted domain configured as pesu.pes.edu | 1: Open /register<br>2: Enter a personal email<br>3: Complete remaining fields<br>4: Click Register | Email arjun.rao@gmail.com, other fields valid | Registration rejected; error "Please register with your institutional email address" | As expected. Automated: test_UT_003_non_institutional_email_is_rejected (identity.test_auth): ok | **Pass** |
| UT_004 | User Registration | FR-001 — Reject a malformed email address | Registration page reachable | 1: Open /register<br>2: Enter an email with no @<br>3: Complete remaining fields<br>4: Click Register | Email arjun.rao.pesu.pes.edu, other fields valid | Registration rejected; error "Enter a valid email address"; no account created | As expected. Automated: test_UT_004_invalid_email_is_rejected (identity.test_auth): ok | **Pass** |
| UT_005 | User Registration | FR-001 — Reject a duplicate institutional email | STU_A already registered | 1: Open /register<br>2: Enter STU_A's email with a fresh roll number | Email stu.a@pesu.pes.edu, roll PES1UG24CS999 | Registration rejected; error "An account with this email already exists"; | As expected. Automated: test_UT_005_duplicate_email_is_rej | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | 3: Click Register | | STU_A unmodified | ected (identity.test_auth): ok | |
| UT_006 | User Registration | FR-001 — Reject a duplicate roll number | STU_A already registered | 1: Open `/register`<br>2: Enter a fresh email with STU_A's roll number<br>3: Click Register | Email `new.student@pesu.pes.edu`, roll `PES1UG24CS001` | Registration rejected; error "This roll number is already registered" | As expected. Automated: `test_UT_006_duplicate_roll_number_is_rejected` (identity.test_auth): ok | **Pass** |
| IT_001 | User Registration | FR-001 — Registration persists a Student row with a hashed password | Test database reachable and inspectable | 1: Register a new Student<br>2: Query the users table for that email<br>3: Inspect the stored password column | Email `arjun.rao@pesu.pes.edu`, password `Str0ng#Pass` | Exactly one row exists; role is `STUDENT`; the password column holds a salted hash; the literal `Str0ng#Pass` appears nowhere in the database or logs | As expected. Automated: `test_IT_001_registration_stores_a_hash_not_the_password` (identity.test_auth): ok | **Pass** |
| IT_002 | User Registration | FR-001 — Two concurrent registrations of the same roll number produce one account | Test database reachable; no account for roll `PES1UG24CS222` | 1: Fire two registration requests with the same roll number simultaneously<br>2: Query the users table | Two requests, both roll `PES1UG24CS222`, different emails | Exactly one account is created; the other is rejected as a duplicate; no partial or orphaned row | As expected. Automated: `test_IT_002_duplicate_roll_number_creates_only_one_account` (identity.test_auth): ok | **Pass** |
| ST_001 | User Registration | FR-001 — End-to-end registration through to a usable account | TEST environment running | 1: Open the portal as an anonymous visitor<br>2: Register with valid details<br>3: Sign in with the new credentials<br>4: Confirm the Student dashboard loads | Name `Arjun Rao`, email `arjun.rao@pesu.pes.edu`, roll `PES1UG24CS111`, password `Str0ng#Pass` | Registration succeeds and the new account can immediately sign in and reach the Student dashboard | As expected. Automated: `test_ST_001_register_sign_in_and_load_student_dashboard` (identity.test_auth): ok | **Pass** |

## UC-02 — Sign in and reach a role-appropriate dashboard

Verifies `FR-002`, `FR-003`, and the sign-in lockout of `NFR-012`.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_007 | Authentication | FR-002 — Sign in with correct credentials | STU_A exists and is active | 1: Open `/login`<br>2: Enter email and | `stu.a@pesu.pes.edu` / `Passw0rd!A` | Sign-in succeeds; a session is established; | As expected. Automated: | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | password<br>3: Click Sign in | | the Student dashboard is displayed | `test_UT_007_correct_credentials_establish_a_session` (identity.test_auth): ok | |
| UT_008 | Authentication | FR-002 — Reject a correct email with a wrong password | STU_A exists and is active | 1: Open `/login`<br>2: Enter the correct email and a wrong password<br>3: Click Sign in | `stu.a@pesu.pes.edu` / `WrongPass1` | Sign-in refused; no session established; message "Invalid email or password" — it does not reveal that the email exists | As expected. Automated: `test_UT_008_wrong_password_is_refused` (identity.test_auth): ok | **Pass** |
| UT_009 | Authentication | FR-002 — Reject an unknown email without disclosing that it is unknown | No account exists for the email used | 1: Open `/login`<br>2: Enter an unregistered email and any password<br>3: Click Sign in | `nobody@pesu.pes.edu` / `AnyPass123` | Sign-in refused with the identical message and a comparable response time to UT_008, so accounts cannot be enumerated | As expected. Automated: `test_UT_009_unknown_email_has_the_same_message` (identity.test_auth): ok | **Pass** |
| UT_010 | Authentication | FR-002 — Reject sign-in with an empty password | STU_A exists | 1: Open `/login`<br>2: Enter a valid email and leave the password empty<br>3: Click Sign in | `stu.a@pesu.pes.edu` / "" (empty) | Sign-in refused; inline error "Password is required"; no authentication attempt is recorded | As expected. Automated: `test_UT_010_empty_password_is_refused` (identity.test_auth): ok | **Pass** |
| UT_011 | Authorization | FR-003 — Every account carries exactly one role | STU_A, FAC_A and ADM_A exist | 1: Query the stored record for each of the three accounts<br>2: Inspect the role field | The three fixture accounts | Each holds exactly one role, drawn from Student, Faculty or Administrator; no account holds none or more than one | As expected. Automated: `test_UT_011_every_account_carries_exactly_one_valid_role` (attempts.test_cohorts): ok | **Pass** |
| IT_003 | Authentication | FR-002 — The session identifier is regenerated on sign-in | STU_A exists; browser developer tools available | 1: Load `/login` and record the pre-authentication session cookie<br>2: Sign in successfully<br>3: Record the cookie again and compare | `stu.a@pesu.pes.edu` / `Passw0rd!A` | The identifier differs before and after; the pre-authentication identifier is invalidated, so a fixated session cannot be reused | As expected. Automated: `test_IT_003_successful_login_rotates_the_session_key` (identity.test_auth): ok | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| IT_004 | Authorization | FR-003 — A Student is refused a Faculty-only API endpoint called directly | STU_A signed in | 1: Sign in as STU_A<br>2: Using the session token, call POST `/api/questions` directly, bypassing the UI<br>3: Inspect the response | Valid Student session token; a well-formed question payload | Response is HTTP 403; no question is created; the body carries no Faculty data | As expected.<br>Automated: `test_IT_004_student_is_refused_faculty_screens_and_endpoints` (identity.tests): ok | **Pass** |
| IT_005 | Authorization | FR-003 — A Faculty member is refused an Administrator-only endpoint | FAC_A signed in | 1: Sign in as FAC_A<br>2: Call POST `/api/admin/courses` with a valid payload<br>3: Inspect the response | Valid Faculty session token; a well-formed course payload | Response is HTTP 403; no course is created | As expected.<br>Automated: `test_IT_005_faculty_is_refused_administrator_screens` (identity.tests): ok | **Pass** |
| IT_006 | Authorization | FR-003 — An unauthenticated client is refused every protected endpoint | Application running; no session held | 1: Enumerate the application's endpoints<br>2: Call each with no session cookie, excluding sign-in and registration<br>3: Record each status code | The full endpoint list | Every protected page redirects to sign-in (HTTP 302 to `/login/?next=…`); every protected API endpoint returns HTTP 401; a view that declares no roles is refused to everyone, so no endpoint defaults to open | As expected.<br>Automated: `test_IT_006_unauthenticated_client_is_refused` (identity.tests): ok | **Pass** |
| IT_038 | Authentication | NFR-012 — Five failed sign-ins within 15 minutes lock the account for 15 minutes | STU_A exists and is active; no failed sign-ins recorded for it | 1: Sign in as STU_A with a wrong password five times<br>2: Sign in with the correct password<br>3: Advance the recorded failures by 16 minutes<br>4: Sign in with the correct password again | `stu.a@pesu.pes.edu` / `WrongPass1` ×5, then `Passw0rd!A` | Step 2 is refused with "Too many failed sign-ins. Try again in 15 minutes." even though the password is right, and no session is established; step 4 succeeds | As expected.<br>Automated: `test_IT_038_lock_lifts_after_15_minutes` (identity.test_lockout): ok<br>`test_IT_038_sixth_attempt_with_right_password_is_refused` (identity.test_lockout): ok | **Pass** |
| ST_002 | Authentication & Authorization | FR-002, FR-003 — Each role lands on its own dashboard and | STU_A, FAC_A, ADM_A exist and are active | 1: Sign in as STU_A, note the landing page, attempt | The three fixture accounts | Each role lands on its own dashboard; every cross-role navigation | As expected.<br>Automated: `test_ST_002_each_role_lands_on_i` | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | cannot reach another's | | /faculty/questions, sign out<br>2: Repeat as FAC_A, attempting /admin/courses<br>3: Repeat as ADM_A, confirming the admin dashboard loads | | is refused with HTTP 403 and an authorisation message; no other role's data is rendered | ts_own_dashboard (identity.tests): ok | |

## UC-03 — Manage courses and assign faculty

Verifies FR-004.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_012 | Course Management | FR-004 — Create a course with code, title and semester | ADM_A signed in | 1: Open /admin/courses<br>2: Click New course<br>3: Enter code, title and semester<br>4: Save | Code UE24CS351A, title Database Management Systems, semester 5 | The course is created and listed as Active; message "Course created" | As expected. Automated: test_UT_012_admin_creates_a_course (identity.test_courses): ok | **Pass** |
| UT_013 | Course Management | FR-004 — Reject a course with any mandatory field empty | ADM_A signed in | 1: Open the New course form<br>2: Leave the Title empty<br>3: Complete code and semester<br>4: Save | Code UE24CS999A, title "" (empty), semester 5 | Creation rejected; inline error "Course title is required"; no course created | As expected. Automated: test_UT_013_empty_course_title_is_rejected (identity.test_courses): ok | **Pass** |
| UT_014 | Course Management | FR-004 — Reject a duplicate code within the same semester | CRS_X exists as UE24CS351A in semester 5 | 1: Open the New course form<br>2: Enter the same code and semester with a different title<br>3: Save | Code UE24CS351A, title DBMS Repeat, semester 5 | Creation rejected; error "A course with this code already exists for the selected semester" | As expected. Automated: test_UT_014_duplicate_code_and_semester_is_rejected (identity.test_courses): ok | **Pass** |
| UT_015 | Course Management | FR-004 — Reject assigning a non-Faculty account as Faculty | CRS_X exists; STU_A is a Student | 1: Open CRS_X<br>2: Open the Assign faculty picker<br>3: Attempt to assign STU_A | STU_A = stu.a@pesu.pes.edu | The assignment is rejected; only Faculty-role accounts are selectable, or the attempt raises "Only Faculty accounts can | As expected. Automated: test_UT_015_student_cannot_be_assigned_as_faculty (identity.test_courses): | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | | | be assigned to a course" | ok | |
| UT_016 | Course Management | FR-004 — Reject a duplicate faculty assignment | FAC_A already assigned to CRS_X | 1: Open CRS_X<br>2: Assign FAC_A again<br>3: Inspect the assigned list | FAC_A on CRS_X | The duplicate is rejected or ignored; FAC_A appears exactly once | As expected. Automated: `test_UT_016_assigning_same_faculty_twice_has_one_assignment` (identity.test_courses): ok | **Pass** |
| IT_007 | Course Management | FR-004 — Archiving preserves history while blocking new work | CRS_X has enrolled Students and one completed examination | 1: As ADM_A, archive CRS_X<br>2: Attempt to create a new examination under it<br>3: Open its past examination's results | CRS_X with EXM_1 completed | New examinations and enrollments are refused with "This course is archived"; the existing examination, attempts and results remain readable | As expected. Automated: `test_IT_007_archived_course_rejects_enrollment` (identity.test_courses): ok | **Pass** |
| IT_008 | Course Management | FR-004 — Assigning a second faculty grants both authoring rights | CRS_X assigned to FAC_A only | 1: As ADM_A, assign FAC_B to CRS_X<br>2: Sign in as FAC_B<br>3: Create a question under CRS_X | FAC_B newly assigned to CRS_X | CRS_X appears in FAC_B's course list and the question is created successfully | As expected. Automated: `test_IT_008_newly_assigned_faculty_sees_course` (identity.test_courses): ok | **Pass** |
| ST_003 | Course Management | FR-004 — End-to-end course setup by an Administrator | TEST environment running | 1: Sign in as ADM_A<br>2: Create a course<br>3: Assign FAC_A to it<br>4: Sign in as FAC_A and confirm the course is listed<br>5: Archive the course as ADM_A<br>6: Confirm FAC_A can no longer add examinations to it | Code UE24CS360A, title Operating Systems, semester 5; FAC_A | Every stage behaves as specified; the archived course remains readable but closed to new work | As expected. Automated: `test_ST_003_admin_creates_and_archives_course` (identity.test_courses): ok | **Pass** |

## UC-04 — Enroll students and gate examination visibility

Verifies FR-005, FR-006.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_017 | Enrollment | FR-005 — Faculty enrolls a Student into their own course | FAC_A signed in and assigned to CRS_X; STU_A exists | 1: Open CRS_X<br>2: Click Enroll student<br>3: Search for STU_A and confirm | STU_A = roll PES1UG24CS001 | STU_A appears on the CRS_X roster; message "Student enrolled" | As expected. Automated: `test_UT_017_faculty_can_enroll_a_student` (identity.test_courses): ok | **Pass** |
| UT_018 | Enrollment | FR-005 — Reject a duplicate enrollment | STU_A already enrolled in CRS_X | 1: Open CRS_X<br>2: Enroll STU_A again<br>3: Inspect the roster | STU_A on CRS_X | The duplicate is rejected with "This student is already enrolled"; the roster shows STU_A once | As expected. Automated: `test_UT_018_duplicate_enrollment_is_rejected` (identity.test_courses): ok | **Pass** |
| UT_019 | Enrollment | FR-005 — Reject enrolling a Faculty or Administrator account | FAC_A signed in on CRS_X | 1: Open the Enroll student picker<br>2: Search for FAC_B and attempt to enroll | FAC_B = `fac.b@pes.edu` | The enrollment is rejected; only Student-role accounts are selectable, or the attempt raises "Only Student accounts can be enrolled" | As expected. Automated: `test_UT_019_faculty_account_cannot_be_enrolled_as_student` (identity.test_courses): ok | **Pass** |
| UT_020 | Enrollment | FR-005 — Faculty cannot enroll into a course they are not assigned to | FAC_B assigned to CRS_Y only | 1: Sign in as FAC_B<br>2: Call the enroll endpoint for CRS_X directly | FAC_B, target CRS_X, student STU_A | The call is refused with HTTP 403; no enrollment is created | As expected. Automated: `test_UT_020_unassigned_faculty_cannot_enroll_students` (identity.test_courses): ok | **Pass** |
| UT_021 | Enrollment | FR-006 — A Student sees only the examinations of enrolled courses | STU_A enrolled in CRS_X only; both CRS_X and CRS_Y have published examinations | 1: Sign in as STU_A<br>2: Open the dashboard<br>3: List the examinations shown | STU_A; CRS_X and CRS_Y each with one published exam | Only the CRS_X examination is listed; the CRS_Y examination does not appear | As expected. Automated: `test_UT_021_student_sees_only_exams_of_enrolled_courses` (attempts.test_cohorts): ok | **Pass** |
| IT_009 | Enrollment | FR-006 — Direct URL access to a non-enrolled examination is refused | STU_A not enrolled in CRS_Y; a CRS_Y examination exists with a known ID | 1: Sign in as STU_A<br>2: Request the CRS_Y examination URL directly<br>3: Inspect the response | STU_A; CRS_Y examination ID | Response is HTTP 403; no title, instructions or questions appear in the body | As expected. Automated: `test_IT_009_admin_can_enroll_a_student` (identity.test_courses): ok | **Pass** |
| IT_010 | Enrollment | FR-006 — The start-attempt endpoint refuses a non-enrolled Student | STU_A not enrolled in CRS_Y; a live CRS_Y examination exists | 1: Sign in as STU_A<br>2: Call POST `/api/attempts` for the CRS_Y examination<br>3: Query the attempts | STU_A; live CRS_Y examination | The call is refused with HTTP 403; no attempt record is created | As expected. Automated: `test_IT_010_draft_exam_is_hidden_from_list` (attempts.test_visibility): ok | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | table | | | `test_IT_010_enrolled_student_sees_published_exam` (attempts.test_visibility): ok `test_IT_010_unenrolled_student_does_not_see_exam` (attempts.test_visibility): ok `test_IT_010_student_not_enrolled_is_refused` (attempts.tests): ok | |
| IT_011 | Enrollment | FR-006 — An unenrolled Student cannot open another course's examination detail | STU_A not enrolled in CRS_Y; a published CRS_Y examination exists | 1: Sign in as STU_A<br>2: Open the CRS_Y examination's detail URL directly | STU_A; published CRS_Y examination | The request is refused with HTTP 403 Forbidden; no examination detail is shown | As expected. Automated: `test_IT_011_unenrolled_student_gets_403_for_exam_detail` (attempts.test_visibility): ok | **Pass** |
| ST_004 | Enrollment | FR-005, FR-006 — End-to-end cohort isolation between two courses | TEST environment; CRS_X and CRS_Y each with a published examination | 1: As FAC_A, enroll STU_A into CRS_X<br>2: As FAC_B, enroll STU_B into CRS_Y<br>3: Sign in as STU_A and list visible examinations<br>4: Sign in as STU_B and do the same<br>5: From each account, request the other's examination URL | STU_A → CRS_X; STU_B → CRS_Y | Each Student sees only their own course's examination; each cross-course URL request is refused with HTTP 403 | As expected. Automated: `test_ST_004_end_to_end_cohort_isolation_between_two_courses` (attempts.test_cohorts): ok | **Pass** |

## UC-05 — Author and manage question bank entries

Verifies `FR-007`, `FR-008`, `FR-009`.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_022 | Question Bank | FR-007 — Create a single-correct MCQ with four options | FAC_A signed in and assigned to CRS_X | 1: Open `/faculty/questions/new`<br>2: Enter the stem | Stem `Which normal form removes transitive dependency?`, | The question is saved to the CRS_X bank and appears in the question list | As expected. Automated: `test_UT_022_valid_single_correct_que` | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | 3: Add four options and mark one correct<br>4: Complete the metadata<br>5: Save | options 1NF/2NF/3NF/BCNF, correct 3NF | | stion_is_saved_and_listed (questionbank.tests): ok | |
| UT_023 | Question Bank | FR-007 — Reject a question with fewer than two options | FAC_A signed in | 1: Open the new question form<br>2: Enter a stem and a single option<br>3: Save | Stem valid, one option 3NF | Creation rejected; error "A question must have between 2 and 6 options" | As expected.<br>Automated: test_UT_023_one_option_is_rejected (questionbank.tests): ok | **Pass** |
| UT_024 | Question Bank | FR-007 — Reject a question with more than six options | FAC_A signed in | 1: Open the new question form<br>2: Enter a stem and seven options<br>3: Save | Stem valid, seven options A–G | Creation rejected with the same boundary message; no question is created | As expected.<br>Automated: test_UT_024_seven_options_are_rejected (questionbank.tests): ok | **Pass** |
| UT_025 | Question Bank | FR-007 — Reject a question with no option marked correct | FAC_A signed in | 1: Enter a stem and four options<br>2: Mark none as correct<br>3: Save | Four options, zero marked correct | Creation rejected; error "Exactly one option must be marked correct" | As expected.<br>Automated: test_UT_025_no_correct_option_is_rejected (questionbank.tests): ok | **Pass** |
| UT_026 | Question Bank | FR-007 — Reject a single-correct question with two options marked correct | FAC_A signed in | 1: Enter a stem and four options<br>2: Mark two as correct<br>3: Save | Four options, 2NF and 3NF both marked correct | Creation rejected with the same message; exactly one correct option is required | As expected.<br>Automated: test_UT_026_two_correct_options_are_rejected (questionbank.tests): ok | **Pass** |
| UT_027 | Question Bank | FR-008 — Reject a question missing any mandatory metadata field | FAC_A signed in | 1: Enter a valid stem and options<br>2: Leave Difficulty unset<br>3: Save; repeat for course, topic, marks and negative marks | Stem and options valid; one metadata field empty each time | Each is rejected with an inline error naming the missing field; no question is created | As expected.<br>Automated: test_UT_027_missing_metadata_is_rejected (questionbank.tests): ok | **Pass** |
| IT_012 | Question Bank | FR-008 — Reject a difficulty value outside the permitted set | FAC_A signed in; API reachable | 1: Submit a question via the API with an invalid difficulty<br>2: Inspect the response | Difficulty Impossible | Rejected with HTTP 400 and "Difficulty must be one of Easy, Medium or Hard" | As expected.<br>Automated: test_IT_012_api_rejects_invalid_difficulty (questionbank.tests): ok | **Pass** |
| IT_013 | Question Bank | FR-009 — A question in a published | Q_SC1 is attached to EXM_1, which is | 1: Sign in as FAC_A<br>2: Open Q_SC1 | Q_SC1, new stem text | The in-place edit is refused; a **new version** | As expected.<br>Automated: | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | examination cannot be edited in place | published | 3: Change the stem<br>4: Save<br>5: Query the question table | | is created instead; the version attached to EXM_1 is byte-identical to before | `test_IT_013_published_question_is_versioned_not_edited` (questionbank.test_versioning): ok | |
| IT_014 | Question Bank | FR-009 — A question that a Student has answered cannot be deleted | Q_SC1 attached to EXM_1; STU_A has answered it in a submitted attempt | 1: Sign in as FAC_A<br>2: Attempt to delete Q_SC1<br>3: Inspect the attempt's stored response | Q_SC1, STU_A's submitted attempt | Deletion refused with "This question has been attempted and cannot be deleted"; the stored response remains resolvable to the original question text | As expected.<br>Automated: `test_IT_014_answered_question_cannot_be_deleted` (questionbank.test_versioning): ok | **Pass** |
| ST_005 | Question Bank | FR-007, FR-008, FR-009 — End-to-end authoring, tagging and version-freeze | TEST environment; FAC_A assigned to CRS_X | 1: Sign in as FAC_A<br>2: Author three tagged questions<br>3: Attach them to a new examination and send it to the Administrator<br>4: Attempt to edit one of the three<br>5: Confirm a new version is created and the sent paper is unchanged | Three MCQs across two topics and two difficulties | Authoring and tagging succeed; once sent (CR-02) or published, the paper's questions are frozen — edits produce versions rather than mutations | As expected.<br>Automated: `test_ST_005_author_tag_publish_then_edits_become_versions` (questionbank.test_versioning): ok | **Pass** |

## UC-06 — Assemble and publish an examination

Verifies FR-010, FR-011, FR-012.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_028 | Exam Authoring | FR-010 — Create an examination with all mandatory attributes | FAC_A signed in and assigned to CRS_X | 1: Open `/faculty/exams/new`<br>2: Enter title, course, duration, declared total marks and instructions<br>3: Save | Title `DBMS Unit Test 1`, course CRS_X, duration `60`, total `50`, instructions free text | The examination is created in **Draft** state and appears in the Faculty examination list | As expected.<br>Automated: `test_UT_028_exam_defaults_to_draft` (examsetup.tests): ok | **Pass** |
| UT_029 | Exam Authoring | FR-010 — Reject a non-positive duration | FAC_A signed in | 1: Open the new examination form<br>2: Enter a duration of zero | Duration `0`, then `-30` | Both rejected with "Duration must be a positive number of minutes"; no | As expected.<br>Automated: `test_UT_029_duration_must_be_positiv` | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | 3: Save; repeat with a negative value | | examination created | e (examsetup.tests): ok | |
| UT_030 | Exam Authoring | FR-010 — Per-examination policy values persist and are read back | A Draft examination exists | 1: Open its settings<br>2: Enable negative marking at 0.5, enable question and option shuffling, set pass mark 40%<br>3: Save and reload the page | Negative `0.5`, shuffle both on, pass mark `40` | All policy values persist exactly as set and are shown unchanged after reload | As expected. Automated: `test_UT_030_exam_policies_are_saved` (examsetup.tests): ok | **Pass** |
| UT_031 | Exam Authoring | FR-010 — Reject a pass mark outside 0–100 | A Draft examination exists | 1: Open its settings<br>2: Enter a pass mark of 150<br>3: Save; repeat with −10 | Pass mark `150`, then `-10` | Both rejected with "Pass mark must be between 0 and 100"; the stored value is unchanged | As expected. Automated: `test_UT_031_pass_mark_must_be_between_zero_and_100` (examsetup.tests): ok | **Pass** |
| UT_032 | Exam Authoring | FR-012 — A Draft examination is invisible to Students | A Draft examination exists on CRS_X; STU_A enrolled in CRS_X | 1: Sign in as STU_A<br>2: Open the dashboard<br>3: Request the Draft examination's URL directly | Draft examination ID | The examination does not appear on the dashboard, and the direct request is refused with HTTP 403 or 404 | As expected. Automated: `test_UT_032_draft_exam_is_invisible_to_students` (examsetup.test_publication): ok | **Pass** |
| UT_033 | Exam Authoring | FR-012 — Sending, then scheduling, moves the examination out of Draft state (CR-02) | A Draft examination with questions attached; ADM_A exists | 1: As FAC_A, open the Draft and click Send to Admin<br>2: Inspect the stored state<br>3: As ADM_A, schedule it<br>4: Inspect the stored state and the audit log | Draft examination with 25 questions | Step 2: SUBMITTED, no publication time. Step 4: PUBLISHED with a server timestamp; audit entries `submit` by FAC_A and `schedule` by ADM_A | As expected. Automated: `test_UT_033_sending_then_scheduling_publishes_the_exam_with_a_server_timestamp` (examsetup.test_publication): ok | **Pass** |
| IT_015 | Exam Authoring | FR-011 — Questions are attached from the bank and stored against the examination | FAC_A signed in; CRS_X bank holds at least ten questions | 1: Open a Draft examination<br>2: Select five questions from the bank<br>3: Save<br>4: Query the examination-question join table | Five known question IDs | Exactly the five selected questions are attached, in the chosen order, with no duplicates and no unselected question present | As expected. Automated: `test_IT_015_exactly_the_selected_questions_attach_in_the_chosen_order` (examsetup.test_publication): ok | **Pass** |
| IT_016 | Exam Authoring | FR-012 — Publishing makes the examination | A Draft examination on CRS_X; STU_A | 1: As FAC_A, send the examination; as ADM_A, | Published examination on | STU_A sees it listed; STU_B does not, since | As expected. Automated: | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | visible to enrolled Students only | enrolled, STU_B not | schedule it<br>2: Sign in as STU_A and check the dashboard<br>3: Sign in as STU_B and check | CRS_X | visibility requires both publication and enrollment | `test_IT_016_published_exam_is_visible_to_enrolled_students_only` (examsetup.test_publication): ok | |
| ST_006 | Exam Authoring | FR-010, FR-011, FR-012 — End-to-end paper assembly and publication | TEST environment; CRS_X bank populated; FAC_A assigned | 1: Sign in as FAC_A<br>2: Create a Draft with title, duration and policy values<br>3: Attach 25 questions from the bank<br>4: Confirm a Student cannot see it<br>5: Send to Admin; as ADM_A, schedule it (CR-02)<br>6: Sign in as an enrolled Student and confirm the paper is listed | 60-minute paper, 25 questions, negative `0.5`, pass mark `40` | The paper is invisible while Draft and while Sent to Admin, becomes visible once the Administrator schedules it, and carries the configured policy values through unchanged. Dashboard transitions are covered by ST_007 | As expected.<br>Automated: `test_ST_006_paper_assembly_and_publication_keep_the_policy_values` (examsetup.test_publication): ok | **Pass** |

## UC-07 — Schedule an examination and surface it on the Student dashboard

Verifies `FR-013`, `FR-014`.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_034 | Scheduling | FR-013 — The Administrator schedules a sent examination with a valid window (CR-02) | An examination of 60 minutes' duration Sent to Admin; ADM_A signed in | 1: Open it from the Examinations queue<br>2: Set window start and end date-times<br>3: Save | Start `2026-09-10 10:00`, end `2026-09-10 12:00` | The schedule is saved and displayed; message "Examination scheduled"; the examination is now Published | As expected.<br>Automated: `test_UT_034_sent_exam_window_is_saved_and_shown` (examsetup.test_schedule): ok | **Pass** |
| UT_035 | Scheduling | FR-013 — Reject a window whose end precedes its start | An examination Sent to Admin; ADM_A signed in | 1: Set an end date-time earlier than the start<br>2: Save | Start `2026-09-10 12:00`, end `2026-09-10 10:00` | Rejected with "The window end must be later than the window start"; no schedule saved and the examination stays unpublished | As expected.<br>Automated: `test_UT_035_window_end_must_be_later_than_start` (examsetup.test_schedule): ok | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_036 | Scheduling | FR-014 — The dashboard classifies an examination as Upcoming before its window | STU_A enrolled in CRS_X; a CRS_X examination scheduled to start tomorrow | 1: Sign in as STU_A<br>2: Open the dashboard<br>3: Locate the examination | Window starting `now + 24 h` | Listed under **Upcoming** with its start date-time; no Start button is offered | As expected.<br>Automated: `test_UT_036_exam_starting_tomorrow_is_upcoming_without_start_button` (identity.test_dashboard): ok | **Pass** |
| UT_037 | Scheduling | FR-014 — The dashboard classifies an examination as Live inside its window | STU_A enrolled; a CRS_X examination whose window is currently open | 1: Sign in as STU_A<br>2: Open the dashboard | Window `now − 10 min` to `now + 50 min` | Listed under **Live** with an active Start button | As expected.<br>Automated: `test_UT_037_open_window_is_live_with_start_button` (identity.test_dashboard): ok | **Pass** |
| UT_038 | Scheduling | FR-014 — The dashboard classifies a submitted examination as Completed | STU_A has one submitted attempt on EXM_1 | 1: Sign in as STU_A<br>2: Open the dashboard | EXM_1 with a submitted attempt | Listed under **Completed**; no Start button is offered | As expected.<br>Automated: `test_UT_038_submitted_exam_is_completed_without_start_button` (identity.test_dashboard): ok | **Pass** |
| IT_017 | Scheduling | FR-013 — The stored window is timezone-unambiguous and server-authoritative | An examination Sent to Admin; database inspectable | 1: As ADM_A, schedule a window from the UI<br>2: Query the stored start and end values<br>3: Change the client machine's timezone and reload | Start `2026-09-10 10:00 IST` | The stored values are unambiguous (UTC or with an explicit offset); the displayed window is unchanged by the client timezone switch | As expected.<br>Automated: `test_IT_017_schedule_is_stored_as_aware_utc_time` (examsetup.test_schedule): ok | **Pass** |
| IT_018 | Scheduling | FR-014 — Dashboard classification is driven by server time, not the client clock | STU_A enrolled; a CRS_X examination whose window opens in one hour | 1: Sign in as STU_A<br>2: Advance the client clock by two hours<br>3: Reload the dashboard | Window opening at `server now + 1 h`; client clock `+2 h` | The examination remains under **Upcoming**; the tampered client clock does not promote it to Live | As expected.<br>Automated: `test_IT_018_classification_uses_server_time_not_client_clock` (identity.test_dashboard): ok | **Pass** |
| ST_007 | Scheduling | FR-013, FR-014 — End-to-end scheduling and dashboard transitions | TEST environment; a published 30-minute examination; STU_A enrolled | 1: As ADM_A, set a 60-minute window opening in two minutes<br>2: As STU_A, confirm it shows Upcoming | Duration 30, window `60 min` | The examination moves Upcoming → Live → Completed on the Student dashboard as server time advances | As expected.<br>Automated: `test_ST_007_exam_moves_upcoming_live_completed_as_serv` | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | 3: Wait for the window to open and refresh<br>4: Sit and submit the paper<br>5: Refresh the dashboard | | | er_time_advances (identity.test_dashboard_flow): ok | |

## UC-08 — Start an attempt and receive the paper

Verifies FR-015, FR-016, FR-017, FR-018.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_039 | Exam Delivery | FR-015 — Starting is refused before the window opens | STU_A enrolled; EXM_1 window opens in ten minutes | 1: Sign in as STU_A<br>2: Attempt to start EXM_1 | Server time window start − 10 min | Refused with "This examination has not opened yet"; no attempt record is created | As expected. Automated: test_UT_039_start_is_refused_before_the_window_opens (attempts.tests): ok | **Pass** |
| UT_040 | Exam Delivery | FR-015 — Starting is refused after the window closes | STU_A enrolled; EXM_1 window closed one minute ago | 1: Sign in as STU_A<br>2: Attempt to start EXM_1 | Server time window end + 1 min | Refused with "This examination has closed"; no attempt record is created | As expected. Automated: test_UT_040_start_is_refused_after_the_window_closes (attempts.tests): ok | **Pass** |
| UT_041 | Exam Delivery | FR-015 — Starting is refused when a submitted attempt already exists | STU_A has one submitted attempt on EXM_1; the window is still open | 1: Sign in as STU_A<br>2: Attempt to start EXM_1 again | STU_A, EXM_1, one submitted attempt | Refused with "You have already submitted this examination"; no second attempt is created | As expected. Automated: test_UT_041_start_is_refused_when_a_submitted_attempt_exists (attempts.tests): ok | **Pass** |
| UT_042 | Exam Delivery | FR-017 — The countdown is computed from server time, not the client clock | STU_A has an attempt in progress on EXM_1, 60-minute duration, 10 minutes elapsed | 1: Note the displayed remaining time<br>2: Set the client clock back by 30 minutes<br>3: Refresh the attempt page<br>4: Note the remaining time again | Elapsed 10 min; client clock moved −30 min | Remaining time reads approximately 50 minutes both times; the client clock change gains the candidate nothing | As expected. Automated: test_UT_042_remaining_time_uses_server_time_for_helper_and_api (attempts.test_timer_payload): ok | **Pass** |
| UT_043 | Exam | FR-018 — The palette | STU_A has an attempt | 1: Start the attempt and | Question 5 across four | The state moves **Not** | As expected. | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | Delivery | shows the correct state for each question | in progress on a 25-question paper | inspect question 5's palette state<br>2: Open question 5, answer nothing, move on, inspect again<br>3: Return and select an option, inspect again<br>4: Mark it for review and inspect again | interactions | **Visited → Not Answered → Answered → Marked for Review**, matching each interaction exactly | Automated: `test_UT_043_palette_moves_through_every_state` (attempts.test_palette): ok | |
| UT_044 | Exam Delivery | FR-018 — Exactly one question is displayed at a time | STU_A has an attempt in progress on a 25-question paper | 1: Load the attempt screen<br>2: Count the question stems rendered in the DOM | 25-question paper | Exactly one stem is rendered; the other 24 are not present in the delivered markup | As expected.<br>Automated: `test_UT_044_exactly_one_stem_is_rendered` (attempts.test_palette): ok | **Pass** |
| IT_019 | Exam Delivery | FR-016 — The attempt record is created and server-stamped before the paper is delivered | STU_A enrolled; EXM_1 window open; network trace available | 1: Click Start<br>2: Capture the request/response sequence<br>3: Query the attempts table | STU_A (`stu.a@pesu.pes.edu` / `Passw0rd!A`, enrolled in CRS_X); EXM_1 ("DBMS Unit Test 1", 60 min, 25 questions, published, window open now) | The attempt row exists with a server-generated start timestamp **before** the question payload is returned; no question data is sent if the row cannot be created | As expected.<br>Automated: `test_IT_019_attempt_row_exists_before_the_paper_is_delivered` (attempts.tests): ok<br>`test_IT_019_no_paper_without_an_attempt_row` (attempts.tests): ok | **Pass** |
| IT_020 | Exam Delivery | FR-016 — The stored start time is the server's, not the client's | STU_A enrolled; EXM_1 window open | 1: Set the client clock forward by one hour<br>2: Start the attempt<br>3: Query the stored start timestamp | Client clock +1 h | The stored start time reflects true server time; the tampered client clock is ignored | As expected.<br>Automated: `test_IT_020_stored_start_time_is_the_servers` (attempts.tests): ok | **Pass** |
| IT_021 | Exam Delivery | FR-018 — Palette state survives navigation and reload | STU_A has an attempt with a mix of palette states across ten questions | 1: Produce all four states across ten questions<br>2: Reload the browser<br>3: Compare every palette state before and after | Ten questions across four states | Every state is identical after reload, because palette state is derived from server-held data rather than browser memory | As expected.<br>Automated: `test_IT_021_states_survive_reload_because_they_come_from_the_server` (attempts.test_palette): ok | **Pass** |
| ST_008 | Exam Delivery | FR-015, FR-016, FR-017, FR-018 — End-to- | TEST environment; EXM_1 live; STU_A | 1: Sign in as STU_A<br>2: Confirm starting is | 60-minute paper, 25 questions | The attempt starts only inside the window, the | As expected.<br>Automated: | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | end paper delivery under a live window | enrolled | refused before the window opens<br>3: Start once the window is open<br>4: Confirm the timer begins at the full duration and counts down<br>5: Navigate across several questions using the palette<br>6: Submit | | server-driven timer counts down correctly, one question shows at a time, and the palette reflects every state transition accurately | `test_ST_008_paper_delivery_under_a_live_window` (attempts.test_attempt_flow): ok | |

## UC-09 — Autosave and resume after interruption

Verifies `FR-019`, `FR-021`.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_045 | Attempt Responses | FR-019 — A response is persisted without any explicit save action | STU_A has an attempt in progress; network trace available | 1: Select an option on question 4<br>2: Take no further action<br>3: Watch the network trace | Question 4, option A | A persistence request is sent automatically within 5 seconds of the selection; no Save button is required or present | As expected. Automated: `test_UT_045_response_is_persisted_with_no_save_button` (attempts.test_autosave): ok | **Pass** |
| UT_046 | Attempt Responses | FR-019 — Changing a response persists the new value, not the old | STU_A has answered question 3 with option B | 1: Return to question 3<br>2: Select option D<br>3: Wait six seconds<br>4: Query the stored response | Question 3, B replaced by D | The stored response reads option D; exactly one response is held for the question | As expected. Automated: `test_UT_046_changing_a_response_keeps_one_row_with_the_new_value` (attempts.test_autosave): ok | **Pass** |
| UT_047 | Attempt Responses | FR-019 — Clearing a response persists the cleared state | STU_A has answered question 3 | 1: Return to question 3<br>2: Click Clear response<br>3: Wait six seconds<br>4: Query the stored response | Question 3, cleared | The stored response is empty and the palette returns the question to **Not Answered** | As expected. Automated: `test_UT_047_clearing_a_response_stores_it_empty` (attempts.test_autosave): ok | **Pass** |
| UT_048 | Attempt Responses | FR-021 — Remaining time on resume is | STU_A started EXM_1 at server time T; 60-minute duration | 1: Note the start time<br>2: Wait 20 minutes<br>3: Reload the attempt | Start T, reload at T + `20 min` | Remaining time reads approximately 40 minutes — derived from | As expected. Automated: `test_UT_048_remaining_time_on_resume_c` | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | computed from the original start | | 4: Read the remaining time | | the original start, not reset by the reload | ounts_from_the_original_start (attempts.test_resume): ok | |
| IT_022 | Attempt Responses | FR-019 — The autosave reaches the server inside the 5-second budget | STU_A has an attempt in progress; server logs timestamped | 1: Select an option and record the client-side timestamp<br>2: Locate the corresponding server-side write<br>3: Compute the difference<br>4: Repeat for ten selections | Ten selections across ten questions | Every response is committed server-side within 5 seconds of selection; the worst observed case is recorded | As expected. Run by hand in Chromium (Playwright) on 3 Nov against build 2d74453: 6 of 6 selections were committed server-side; worst case 119 ms, mean 117 ms from selection to server response (budget 5 000 ms) | **Pass** |
| IT_023 | Attempt Responses | FR-019 — A browser crash loses no response already autosaved | STU_A has an attempt in progress with six questions answered | 1: Answer six questions<br>2: Wait six seconds<br>3: Kill the browser process without submitting<br>4: Reopen, sign in and resume | Six answered questions | All six responses are restored; nothing is lost, because persistence did not depend on submission | As expected. Automated: test_IT_023_saved_responses_survive_a_browser_that_never_submits (attempts.test_autosave): ok | **Pass** |
| IT_024 | Attempt Responses | FR-021 — Resuming restores responses and does not grant extra time | STU_A started EXM_1 at T; answered four questions by T + 20 min | 1: Answer four questions<br>2: Disconnect the network for five minutes<br>3: Reconnect and reopen the attempt<br>4: Inspect the restored answers and remaining time | Start T, interruption T+20 to T+25 | All four answers are restored; remaining time reads approximately 35 minutes, so the interruption grants no extra time | As expected. Automated: test_IT_024_resume_restores_answers_and_grants_no_extra_time (attempts.test_resume): ok | **Pass** |
| IT_025 | Attempt Responses | FR-021 — Resuming on a different device restores the same attempt state | STU_A has an attempt in progress on profile A with three answers | 1: Answer three questions on profile A<br>2: Sign in as STU_A on profile B<br>3: Open the attempt | Three answered questions; two profiles | Profile B resumes the same attempt with the three answers present and time still measured from the original start | As expected. Automated: test_IT_025_another_device_resumes_the_same_attempt (attempts.test_resume): ok | **Pass** |
| ST_009 | Attempt Responses | FR-019, FR-021 — End-to-end answer, disconnect, resume and submit | TEST environment; EXM_1 live; STU_A enrolled | 1: Start EXM_1<br>2: Answer five questions, changing two and clearing one<br>3: Disconnect for three minutes<br>4: Reconnect and resume | 25-question paper; five answered, two changed, one cleared | Every selection, change and clear survives the interruption exactly; remaining time reflects the original start; the submitted attempt records precisely the | As expected. Automated: test_ST_009_answer_change_clear_resume_and_submit (attempts.test_attempt_flow): ok | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | 5: Verify all responses and remaining time<br>6: Answer the rest and submit<br>7: Inspect the stored attempt | | final state of every question | | |

## UC-10 — Automatic submission on timer expiry

Verifies FR-020.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_049 | Auto-Submission | FR-020 — The attempt submits itself when remaining time reaches zero | STU_A has an attempt in progress with 30 seconds remaining | 1: Answer one more question<br>2: Take no action and let the timer run to zero<br>3: Observe the screen | Remaining `30 s`, then `0` | At zero the attempt is submitted automatically without any click; the Student sees "Time is up. Your paper has been submitted." | As expected. Automated: `test_UT_049_the_paper_submits_itself_at_zero` (attempts.test_auto_submit): ok | **Pass** |
| UT_050 | Auto-Submission | FR-020 — Auto-submission retains every response persisted up to that instant | STU_A has an attempt with eight answered questions and 20 seconds remaining | 1: Confirm eight answers<br>2: Let the timer expire<br>3: Query the submitted attempt's responses | Eight answered of 25 | All eight responses are present; the remaining 17 are recorded as unattempted | As expected. Automated: `test_UT_050_every_response_saved_before_expiry_is_kept` (attempts.test_auto_submit): ok | **Pass** |
| UT_051 | Auto-Submission | FR-020 — No further response is accepted after auto-submission | STU_A's attempt has just auto-submitted | 1: Attempt to select an option on the last displayed question<br>2: Call the response-save endpoint directly with the old attempt ID | Auto-submitted attempt ID | The UI is read-only; the direct API call is refused with "This attempt has already been submitted"; the stored responses are unchanged | As expected. Automated: `test_UT_051_no_response_is_accepted_after_auto_submission` (attempts.test_auto_submit): ok | **Pass** |
| UT_052 | Auto-Submission | FR-020 — The attempt is recorded as submitted with a server timestamp | An attempt that has just auto-submitted | 1: Query the attempt record<br>2: Inspect its submission timestamp | Auto-submitted attempt | The submission timestamp (`submitted_at`) is set, which is what marks the attempt | As expected. Automated: `test_UT_052_submitted_at_is_the_start_plus_the_d` | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | | | submitted; it is server-generated and equals the attempt start plus the configured duration | uration (attempts.test_auto_submit): ok | |
| IT_026 | Auto-Submission | FR-020 — Expiry is enforced server-side even if the browser is closed | STU_A has an attempt in progress with one minute remaining | 1: Answer two questions<br>2: Close the browser entirely<br>3: Wait until after the duration elapses<br>4: Sign in again and open the examination<br>5: Query the attempt record | Browser closed at T − 60 s | The attempt is recorded as submitted with the two answers retained; expiry did not depend on the browser being open | As expected. Automated: test_IT_026_expiry_does_not_need_the_browser_open (attempts.test_auto_submit): ok | **Pass** |
| IT_027 | Auto-Submission | FR-020 — A client-side clock change cannot defer auto-submission | STU_A has an attempt with two minutes remaining | 1: Set the client clock back by one hour<br>2: Let real time pass beyond the true expiry<br>3: Attempt to answer a question | Client clock −1 h; real elapsed time beyond duration | The attempt is auto-submitted at the true server expiry; further answers are refused | As expected. Automated: test_IT_027_a_client_clock_cannot_defer_expiry (attempts.test_auto_submit): ok | **Pass** |
| IT_028 | Auto-Submission | FR-020, FR-022 — An auto-submitted attempt enters evaluation like a manual one | Auto-submission has just occurred for STU_A on EXM_1 | 1: Let the attempt auto-submit<br>2: Query the evaluation record within 5 seconds | Auto-submitted attempt on a 25-question paper | The attempt is evaluated automatically with a total score, percentage and Pass/Fail outcome, identical in treatment to a manually submitted attempt | As expected. Automated: test_IT_028_an_auto_submitted_attempt_is_evaluated (attempts.test_auto_submit): ok | **Pass** |
| ST_010 | Auto-Submission | FR-020 — End-to-end timed expiry on a short paper | TEST environment; a 3-minute examination with five questions; STU_A enrolled | 1: Start the examination<br>2: Answer three of the five questions<br>3: Let the full three minutes elapse without submitting<br>4: Observe the screen at zero<br>5: Sign in as FAC_A and inspect the | 3-minute paper, five questions, three answered | The paper submits itself exactly at expiry, the three answers are retained, the two unanswered are recorded as unattempted, and Faculty sees a normally submitted attempt | As expected. Automated: test_ST_010_three_minute_paper_end_to_end (attempts.test_auto_submit): ok | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | attempt | | | | |

## UC-11 — Evaluate a submitted attempt automatically

Verifies FR-022, FR-023, FR-024.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_053 | Evaluation | FR-023 — Award full positive marks for a correct response | Scoring engine callable in isolation; Q_SC1 = +2 / −0.5, correct option B | 1: Invoke the evaluator with a response of option B<br>2: Read the awarded marks | Response B, correct B, positive 2, negative 0.5 | The question is awarded exactly +2.0 | As expected. Automated: test_UT_053_correct_response_earns_full_positive_marks (evaluation.tests): ok | **Pass** |
| UT_054 | Evaluation | FR-023 — Deduct the configured negative marks for an incorrect response | Scoring engine callable; negative marking enabled at 0.5 | 1: Invoke the evaluator with a response of option A<br>2: Read the awarded marks | Response A, correct B, positive 2, negative 0.5 | The question is awarded exactly −0.5 | As expected. Automated: test_UT_054_incorrect_response_loses_the_configured_negative_marks (evaluation.tests): ok | **Pass** |
| UT_055 | Evaluation | FR-023 — Award zero and apply no deduction to an unattempted question | Scoring engine callable; negative marking enabled at 0.5 | 1: Invoke the evaluator with no response recorded<br>2: Read the awarded marks | Response null, correct B, negative 0.5 | The question is awarded exactly 0.0; the penalty is **not** applied — the boundary that separates skipping from guessing | As expected. Automated: test_UT_055_unanswered_question_earns_zero_with_no_deduction (evaluation.tests): ok | **Pass** |
| UT_056 | Evaluation | FR-023 — Apply no deduction when negative marking is disabled | Scoring engine callable; negative marking disabled on the examination | 1: Invoke the evaluator with an incorrect response<br>2: Read the awarded marks | Response A, correct B, negative marking off | The question is awarded exactly 0.0, not a negative value | As expected. Automated: test_UT_056_no_deduction_when_negative_marking_is_disabled (evaluation.tests): ok | **Pass** |
| UT_057 | Evaluation | FR-024 — Compute total, percentage and a Pass outcome at the pass mark | Scoring engine callable; pass mark 40% | 1: Evaluate an attempt scoring 20 of 50<br>2: Read total, percentage and outcome | Total 20, declared total 50, pass mark 40% | Total 20, percentage 40.0%, outcome **Pass** — the boundary is inclusive | As expected. Automated: test_UT_057_score_exactly_at_the_pass_mark_is_a_pass (evaluation.tests): ok | **Pass** |
| UT_058 | Evaluation | FR-024 — Compute a Fail outcome just below the pass mark | Scoring engine callable; pass mark 40% | 1: Evaluate an attempt scoring 19.5 of 50<br>2: Read the outcome | Total 19.5, declared total 50, pass mark 40% | Percentage 39.0%, outcome **Fail** | As expected. Automated: test_UT_058_score_just_below_the_pass_mark_is_a_fail (evaluation.tests): ok | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| IT_029 | Evaluation | FR-022 — Every submitted attempt is evaluated without human intervention | EXM_1 with three submitted attempts | 1: Submit three attempts<br>2: Take no Faculty action of any kind<br>3: Query the evaluation records | Three submitted attempts | All three carry a computed score, percentage and outcome; none awaits a manual trigger or approval | As expected. Automated: `test_IT_029_every_submitted_attempt_is_evaluated_without_faculty_action` (evaluation.tests): ok | **Pass** |
| IT_030 | Evaluation | FR-022, NFR-002 — A 100-question attempt is evaluated within 5 seconds | An examination of 100 questions with a submitted attempt; server timestamps available | 1: Submit and record the server submission timestamp<br>2: Poll for the evaluation record and record its completion timestamp<br>3: Compute the elapsed time<br>4: Repeat over ten attempts | 100-question paper, ten attempts | Every evaluation completes within 5 seconds of submission; the worst observed elapsed time is recorded | As expected. Automated: `test_IT_030_NFR_002_100_question_attempt_is_evaluated_within_5_seconds` (evaluation.test_performance): ok | **Pass** |
| IT_031 | Evaluation | FR-023, FR-024 — A mixed attempt is scored correctly end to end | An examination of 10 questions at +2 / −0.5 with a submitted attempt | 1: Submit an attempt with 6 correct, 3 incorrect and 1 unattempted<br>2: Query the computed total, percentage and outcome | 6 correct, 3 incorrect, 1 skipped; +2 / −0.5; declared total 20; pass mark 40% | Total is `(6 × 2) − (3 × 0.5) − 0 = 10.5`; percentage 52.5%; outcome **Pass**; the skipped question contributes exactly zero | As expected. Automated: `test_IT_031_mixed_attempt_is_scored_end_to_end` (evaluation.tests): ok | **Pass** |
| ST_011 | Evaluation | FR-022, FR-023, FR-024 — End-to-end automatic scoring of a full cohort | TEST environment; EXM_1 with 25 questions; five enrolled Students | 1: Have five Students sit and submit EXM_1 with differing answer patterns<br>2: Take no Faculty action<br>3: Inspect each attempt's score, percentage and outcome<br>4: Hand-compute two of the five from the answer key and compare | Five attempts across a 25-question paper | All five are evaluated automatically; the two hand-computed totals match the system's exactly; Pass/Fail is correct against the configured pass mark | As expected. Automated: `test_ST_011_five_students_are_scored_with_no_faculty_action` (evaluation.test_cohort): ok | **Pass** |

## UC-12 — Publish results and display the scorecard

Verifies FR-025, FR-026, FR-027.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_059 | Results | FR-025 — An | STU_A has a submitted, | 1: Sign in as STU_A | Evaluated but | No score, percentage or | As expected. Automated: | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | evaluated result is withheld until Faculty publishes | evaluated attempt on EXM_1; results not published | 2: Open the dashboard and the examination<br>3: Look for a score | unpublished result | outcome is shown; the examination reads "Results not yet published" | `test_UT_059_unpublished_result_is_withheld` (results.test_publish): ok | |
| UT_060 | Results | FR-026 — Publishing makes the result visible to the Student, and only once the window has closed | The situation of UT_059: results unpublished | 1: Sign in as FAC_A<br>2: While the window of EXM_1 is still open, try to publish its results<br>3: After the window closes, publish the results<br>4: Sign in as STU_A and reopen the examination | EXM_1 results published | Step 2 is refused and the results stay unpublished; step 3 succeeds and the scorecard is now visible with total, percentage and outcome | As expected. Automated: `test_UT_060_publishing_makes_the_result_visible` (results.test_publish): ok | **Pass** |
| UT_061 | Results | FR-026 — Unpublishing withdraws a previously visible result | EXM_1 results published and seen by STU_A | 1: Sign in as FAC_A<br>2: Unpublish the results<br>3: Sign in as STU_A and reopen the examination | EXM_1 results unpublished | The scorecard is no longer shown; the examination reverts to "Results not yet published"; the stored score is unchanged | As expected. Automated: `test_UT_061_unpublishing_withdraws_it_and_keeps_the_score` (results.test_publish): ok | **Pass** |
| UT_062 | Results | FR-027 — The scorecard shows total, percentage, outcome and per-question marks | EXM_1 results published; STU_A scored 32 of 50 | 1: Sign in as STU_A<br>2: Open the scorecard<br>3: Inspect every displayed field | Total 32, declared 50, pass mark 40%, 25 questions | Total 32, percentage `64.0%`, outcome **Pass**, and a per-question line for all 25 questions showing the marks obtained on each | As expected. Automated: `test_UT_062_scorecard_shows_total_percentage_outcome_and_every_question` (results.test_scorecard): ok | **Pass** |
| UT_063 | Results | FR-027 — Per-question marks on the scorecard sum to the reported total | EXM_1 results published for STU_A | 1: Open the scorecard<br>2: Sum the per-question marks<br>3: Compare with the displayed total | 25 per-question values including negatives | The sum equals the displayed total exactly | As expected. Automated: `test_UT_063_per_question_marks_sum_to_the_total` (results.test_scorecard): ok | **Pass** |
| IT_032 | Results | FR-025 — An unpublished result is unreachable through the API, not merely hidden in the UI | STU_A has an evaluated but unpublished result | 1: Sign in as STU_A<br>2: Call GET `/api/results/{attemptId}` directly<br>3: Inspect the response body | STU_A's attempt ID; results unpublished | Refused with HTTP 403; no score, percentage or per-question mark appears anywhere in the body | As expected. Automated: `test_IT_032_unpublished_result_is_403_through_the_api` (results.test_publish): ok | **Pass** |
| IT_033 | Results | FR-025 — One Student cannot read another's result | EXM_1 results published; STU_A and STU_B both attempted | 1: Sign in as STU_A<br>2: Call the result endpoint with STU_B's attempt ID | STU_B's attempt ID | Refused with HTTP 403; STU_B's score is not disclosed | As expected. Automated: `test_IT_033_another_students_result_is_403` (results.test_publish): ok | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| IT_034 | Results | FR-026, FR-027 — The published scorecard matches the stored evaluation exactly | EXM_1 sat by five Students and published | 1: Publish results<br>2: For each Student, open the scorecard<br>3: Compare each displayed total, percentage and outcome against the stored evaluation record | Five evaluated attempts | Every displayed figure matches its stored evaluation exactly; no rounding or transcription discrepancy | As expected. Automated: `test_IT_034_every_published_scorecard_matches_the_stored_evaluation` (results.test_scorecard): ok | **Pass** |
| ST_012 | Results | FR-025, FR-026, FR-027 — End-to-end publication and scorecard | TEST environment; EXM_1 sat by five Students | 1: Confirm no Student can see a score before publication<br>2: As FAC_A, publish<br>3: As each Student, open the scorecard and verify the figures<br>4: Unpublish and confirm the scorecards disappear | Five evaluated attempts | Results stay hidden until published, every scorecard matches the stored evaluation, and unpublishing withdraws visibility without altering the data | As expected. Automated: `test_ST_012_end_to_end_publication_and_scorecard` (results.test_scorecard): ok | **Pass** |

## UC-13 — Log integrity events and maintain the audit trail

Verifies FR-028, FR-029, FR-030.

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| UT_064 | Integrity | FR-028 — A single focus-loss event is recorded with a timestamp | STU_A has an attempt in progress on EXM_1 | 1: Switch to another browser tab once<br>2: Return to the attempt<br>3: Query the integrity event log for the attempt | One tab switch | Exactly one focus-loss event is recorded against the attempt, carrying a server-side timestamp | As expected. Automated: `test_UT_064_one_focus_loss_is_one_event_with_a_server_time` (integrity.test_focus): ok | **Pass** |
| UT_065 | Integrity | FR-028 — Repeated focus losses are each recorded separately | STU_A has an attempt in progress | 1: Switch away and back three times<br>2: Query the event log | Three tab switches | Exactly three events are recorded, each with its own distinct timestamp; none is coalesced or dropped | As expected. Automated: `test_UT_065_three_losses_are_three_events` (integrity.test_focus): ok | **Pass** |
| UT_066 | Integrity | FR-028 — No event is recorded when focus is never lost | STU_A has an attempt in progress | 1: Complete the attempt without leaving the window<br>2: Query the event log | Zero tab switches | The event log for the attempt is empty; the detector does not fire on ordinary in-page interaction | As expected. Automated: `test_UT_066_no_loss_means_no_event` (integrity.test_focus): ok | **Pass** |
| UT_067 | Integrity | FR-028 — The recorded timestamp is server- | STU_A has an attempt in progress | 1: Set the client clock forward by one hour | Client clock +1 h; one tab switch | The stored timestamp reflects | As expected. Automated: `test_UT_067_client_s` | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | side, not client-supplied | | 2: Switch tabs once<br>3: Query the recorded event timestamp | | true server time, not the tampered client clock | `upplied_time_is_ignored` (integrity.test_focus): ok | |
| UT_068 | Audit | FR-030 — An audit entry records actor, action, entity, timestamp and changed values | FAC_A signed in; audit log inspectable | 1: Edit a question's stem and its answer key<br>2: Query the audit log for that question | Question Q_SC1, stem and correct option changed | One entry exists carrying the actor FAC_A, the action, the affected entity, a server timestamp, and both the previous and the new stem and answer key | As expected. Automated: `test_UT_068_an_edit_records_old_and_new_values` (repository.test_audit): ok `test_UT_068_question_edit_records_old_and_new_stem_and_answer_key` (repository.test_audit_coverage): ok | **Pass** |
| IT_035 | Integrity | FR-029 — No correct-answer data reaches the candidate during an active attempt | STU_A has an attempt in progress on EXM_1 | 1: Capture every HTTP response delivered to the browser during the attempt<br>2: Search the payloads for `isCorrect`, `correctOption`, `answerKey` and the known correct values<br>3: Inspect the rendered HTML and client-side JavaScript state | EXM_1; known answer key | No response body, HTML attribute or client-side variable exposes which option is correct; the key is absent from everything the browser receives | As expected. Automated: `test_IT_035_paper_payload_never_contains_correct_answer_data` (attempts.test_timer_payload): ok | **Pass** |
| IT_036 | Audit | FR-030 — Every mutation class writes an audit entry | ADM_A and FAC_A signed in | 1: Create a user, a course, a question and an examination, and submit an attempt<br>2: Query the audit log for each<br>3: Check who is recorded for that attempt's submission and evaluation<br>4: Change several examination fields at once<br>5: Publish results | One mutation of each of the five entity classes: User, Course, Question, Exam, Attempt | An entry exists for every one of the five, so no mutation class is silently unlogged; the submission is attributed to the Student and the evaluation to the system; every changed examination field is recorded; the results publication is recorded | As expected. Automated: `test_IT_036_every_mutation_class_writes_an_entry` (repository.test_audit): ok | **Pass** |
| IT_037 | Audit | FR-030 — The audit log is append-only and resists modification | Audit entries exist; database reachable | 1: Attempt to update an existing audit row through the application | An existing audit entry | Both attempts are refused; the application exposes | As expected. Automated: `test_IT_037_audit_rows_cannot_be_changed` | **Pass** |

| Test Case ID | Name of Module | Test case description | Pre-conditions | Test Steps | Test data | Expected Results | Actual Result | Test Result |
|---|---|---|---|---|---|---|---|---|
| | | | | 2: Attempt to delete one<br>3: Inspect the table's constraints | | no update or delete path for audit rows; the entry is unchanged | `_or_deleted` (repository.test_audit): ok | |
| ST_013 | Integrity & Audit | FR-028, FR-029, FR-030 — End-to-end evidentiary trail across an examination cycle | TEST environment; `EXM_1` live; `STU_A` enrolled | 1: Sit `EXM_1` as `STU_A`, switching tabs twice<br>2: Confirm no answer key appears in any captured payload<br>3: Submit<br>4: As `FAC_A`, publish results<br>5: As `ADM_A`, open the audit log filtered to `EXM_1` and reconcile it against everything that occurred | One attempt, two focus-loss events, one publication | The log accounts for every state change — attempt creation, submission, evaluation, publication — the two integrity events are attached to the attempt for Faculty review, and no answer key ever reached the candidate | Automated: `test_ST_013_admin_reads_the_log_filtered_to_one_exam` (repository.test_audit): ok. Browser re-run (System Validation Report, 3 Nov): the audit log accounts for creation, submission, evaluation and publication, and no answer key reached the browser, but the attempt carries **6 focus-loss events for 2 tab switches**: every question page left adds one (Defect 1). After the fix (11 Nov, D-15), browser run 3: **2 focus-loss events for 2 tab switches**, still 2 after answering and submitting; audit log and payloads as before | **Pass** |